

What Not To Miss While DEVELOPING WEARABLE TECH

When wearables first came out, fitness trackers and smartwatches dominated the market. With so many fresh and inventive products being developed, this is no longer the case. As complexities and user requirements increase, the most pressing decision for any designer is the selection of a controller

AARYAA
PADHYEGURJAR

Simply speaking, a wearable is any device users can place on their bodies. These frequently function as replacement for or integration with already-existing accessories like watches. Thanks to the Internet of Things (IoT) technologies this market sector is expanding at a meteoric rate. In addition to the consumer market, the medical sector is driving demand for gadgets that track bodily processes and diseases. From smartwatches, bands, glasses, and rings to e-textiles, tattoos, and stickers—all come under the category of wearables.

Selecting the right controller

The microcontroller is the most crucial electronic part of wearable technology. It must be able to address all requirements of a wearable device, the most important ones being low power consumption, ability to communicate wirelessly, and compact size. The controller frequently needs to support a variety of wireless protocols, including Wi-Fi, ANT+, Bluetooth Low Energy (BLE), and proprietary IEEE 802.15.4 based protocols. In some cases, the equipment has to handle multiple protocols.

Microcontrollers that can combine the majority of functions into a single chip may support the majority of wearable devices. In general, the 32-bit ARM architecture offers good

The first wearable

The first wearable technology with a significant mass market influence is thought to be the Hewlett Packard HP-01. Although it was marketed as a calculator wristwatch, it also displayed other things, including the day of the week, alarm, timer, stopwatch, calendar, and time of day. Using the watch, one could know the time, figure out how much a long-distance call would cost, calculate distances, and much more on the go. All these features were available as early as 1977. The price of the watch in today's money was almost \$3000.

performance and energy efficiency. Still, it pretty much depends on your needs and requirements. Here are some things you need to keep in mind while selecting the right chip for your application.

Size and form factor. Many common controllers for wearable devices are quite compact, and these need to be in order to conveniently attach to a wearable. The controllers must, however, simultaneously include more features into the same area. The minimal feature size of the technique used to fabricate the silicon system on chip (SoC) is a crucial aspect to pay attention to. The smaller the feature size of the silicon wafer, the denser the chip will be. Therefore, more features and functionalities can be crammed onto a single chip.

Low power consumption. Reducing the power usage of wearable technology presents particular difficulties because many gadgets are battery-powered. As majority of wearable gadgets are monitoring devices, they must always be on,



The first wearable: HP-01 (Credit: The Museum of HP Calculators)